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MSCF Probability, August 2018: Homework Problems, Set 4

1. Let $X | P = p \sim \text{Bin}(n, p)$ and $P \sim \text{Beta}(\alpha, \beta)$.
 - (a) Find the marginal distribution of X .
 - (b) Derive $f_{P|X}(p | x)$, and show that $P|X = x \sim \text{Beta}(\alpha + x, \beta + n - x)$ when $x = 0, 1, \dots, n$.
 - (c) **Application:**
 The following example has the same setup as this problem (with specific n, α and β), and makes use of the result that was obtained in part (b).
 A credit consultant is responsible for the lines of credit of ten large companies. The probability that any given company of the ten will use its line of credit in the coming month is P , independent of the other companies in the portfolio. However, the value of P is unknown. The credit consultant decides to model P as $\text{Beta}(3, 2)$. In the first month, four of the ten companies used their line of credit.
 - i. Based on the first month's experience, what model should the credit consultant use for P (instead of $\text{Beta}(3, 2)$) in the second month? (In other words, what is the distribution of $P|X = 4$?)
 - ii. Based on the first month experience, what is the probability that more than 3 companies will use their line of credit in the second month?**Guidance:** Using the updated model for P you found in (i), find $f_X(x)$ (using the general density you found in (a)) and then calculate $P(X > 3)$
2. The proportion of defective products coming out of a production line, P , is a random variable, and varies daily according to a beta distribution with parameters $\alpha = 1$ and $\beta = 9$. On a randomly chosen day, 100 products are sampled and inspected. Let X be the number of defective products out of 100. Find $E(X)$.
3. Suppose that X_1, X_2, X_3 have the following joint pdf:

$$f_{X_1 X_2 X_3}(x_1, x_2, x_3) = \begin{cases} 6e^{-(x_1 + 2x_2 + 3x_3)} & x_i > 0, \quad i = 1, 2, 3 \\ 0 & \text{otherwise} \end{cases}$$

Are X_1, X_2 , and X_3 independent? Justify your answer rigorously.

4. Suppose that X and Y are independent, with X having the $\text{Exponential}(\lambda_1)$ distribution, and Y having the $\text{Exponential}(\lambda_2)$ distribution. Does $X + Y$ also have the exponential distribution? Justify your answer rigorously.
5. Consider the following situation:
 - $Y \sim N(\mu, \sigma^2)$
 - $(X | Y = y) \sim N(y, \tau^2)$ [FYI: τ is the Greek letter Tau]

We would like to find the (marginal) distribution of X .

One strategy would be to find $f_{XY}(x, y)$ and then use the “Law of Total Probability” for distributions that we talked about in class:

$$f_X(x) = \int_y \underbrace{f_{X|Y}(x | y) \cdot f_Y(y)}_{f_{XY}(x, y)} dy$$

While this is “doable”, it requires some algebraic manipulations that may not be obvious. Another strategy is to find the MGF $M_X(t)$ and use it to identify the distribution of X . To that end we use:

$$M_X(t) = E(e^{tX}) = E[E(e^{tX}|Y)] = \dots$$

where the “inner” expected value, $E(e^{tX}|Y)$ is the moment generating function of the random variable $X|Y = y$. **Find $M_X(t)$ and use to determine the distribution of X .**

$$\begin{aligned}
 1. (a) \quad f_X(x) &= \int_0^1 f_{X|P}(x|p) \cdot f_P(p) dp \\
 &= \int_0^1 \binom{n}{x} p^x (1-p)^{n-x} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1-p)^{\beta-1} dp \\
 &= \binom{n}{x} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 p^{\alpha+x-1} (1-p)^{n+\beta-x-1} dp \\
 &= \binom{n}{x} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+x)\Gamma(n+\beta-x)}{\Gamma(\alpha+n+\beta)}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad f_{P|X}(p|x) &= \frac{f_{X|P}(x|p) f_P(p)}{f_X(x)} \\
 &= \frac{\binom{n}{x} p^x (1-p)^{n-x} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} p^{\alpha-1} (1-p)^{\beta-1}}{\binom{n}{x} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+x)\Gamma(n+\beta-x)}{\Gamma(\alpha+n+\beta)}} \\
 &= \frac{\Gamma(\alpha+n+\beta)}{\Gamma(\alpha+x)\Gamma(n+\beta-x)} \cdot p^{\alpha+x-1} (1-p)^{n+\beta-x-1}
 \end{aligned}$$

$$\therefore P|X=x \sim \text{Beta}(\alpha+x, n+\beta-x) \quad \square$$

(c) $n=10$ $P \sim \text{Beta}(3, 2)$ $x=4$ in the first month

(i) $P|X=x \sim \text{Beta}(\alpha+x, n+\beta-x)$
 $\alpha+x = 4+3 = 7$
 $n+\beta-x = 10+2-4 = 8$

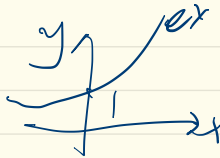
$$\therefore P|X \sim \text{Beta}(7, 8)$$

(ii) $P \sim \text{Beta}(7, 8)$
 From (a) $f_X(x) = \binom{n}{x} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+x)\Gamma(\beta-x)}{\Gamma(\alpha+\beta)}$
 $\therefore f_X(x) = \binom{10}{x} \frac{\Gamma(15)}{\Gamma(7)\Gamma(8)} \cdot \frac{\Gamma(7+x)\Gamma(8-x)}{\Gamma(15)}$

$$\begin{aligned} P(X > 3) &= 1 - P(X=0) - P(X=1) - P(X=2) - P(X=3) \\ &= 1 - \frac{\Gamma(15)}{\Gamma(7)\Gamma(8)\Gamma(15)} \left[\binom{10}{0}\Gamma(7)\Gamma(8) + \binom{10}{1}\Gamma(8)\Gamma(7) \right. \\ &\quad \left. + \binom{10}{2}\Gamma(9)\Gamma(6) + \binom{10}{3}\Gamma(10)\Gamma(5) \right] \\ &= 1 - \frac{4303}{14858} \\ &= 0.71039 \end{aligned}$$

2. $X = \#$ defective product in the sample.
 $E(X) = E(E(X|P))$, where $E(X|P=p) = 100p$.
 $= E(100P)$, where P is a RV, $P \sim \text{Beta}(1, 9)$
 $= 100 E(P)$
 $= 100 \cdot \frac{\alpha}{\alpha+\beta}$
 $= 100 \cdot \frac{1}{10}$
 $= 10$

3. $f_{X_1} = \int_0^\infty \int_{x_1 x_2}^\infty f_{X_1 X_2 X_3} dx_2 dx_3$



$$\begin{aligned}
 &= 6 \int_0^\infty e^{-x_1} \cdot e^{-2x_1} e^{-3x_1} dx_2 dx_3 \\
 &= 6e^{-x_1} \frac{1}{2} \int_0^\infty (e^{-2x_1}) e^{-3x_1} dx_3 \\
 &= 3e^{-x_1} \int_0^\infty e^{-2x_3} dx_3 \\
 &= e^{-x_1}, \quad x_1 > 0.
 \end{aligned}$$

By symmetry, $f_{X_2} = 2e^{-2x_2} (x_2 > 0)$; $f_{X_3} = 3e^{-3x_3} (x_3 > 0)$

Also, $f_{X_1}, f_{X_2}, f_{X_3} > 0$.

$$\begin{aligned}
 f_{X_1}(x_1) f_{X_2}(x_2) f_{X_3}(x_3) &= (e^{-x_1}) (2e^{-2x_2}) (3e^{-3x_3}) = 6e^{-(x_1 + 2x_2 + 3x_3)} \\
 &= f_{X_1 X_2 X_3}(x_1, x_2, x_3) \quad \forall x_i > 0.
 \end{aligned}$$

$\therefore x_1, x_2, x_3$ are independent.

4. x, y , indep
 $x \sim \exp(\lambda_1)$
 $y \sim \exp(\lambda_2)$

Since x and y are independent, let $z = x + y$.

$$\begin{aligned}
 M_z(t) &= M_x(t) \cdot M_y(t) \\
 &= \frac{\lambda_1}{\lambda_1 - t} \cdot \frac{\lambda_2}{\lambda_2 - t} \\
 &= \frac{\lambda_1 \lambda_2}{\lambda_1 \lambda_2 - (\lambda_1 + \lambda_2)t + t^2}
 \end{aligned}$$

Since there's a t^2 in denominator, there's no way for $M_z(t)$ to have a form of exponential distribution's MGF.

$\therefore x + y$ does not have exponential distribution

$$5. \quad Y \sim N(\mu, \sigma^2)$$

$$(X|Y=y) \sim N(y, \tau^2)$$

$$M_X(t) = E(e^{tX})$$

$$= E(E(e^{tX}|Y))$$

$$= E(\exp(Yt + \frac{\tau^2 t^2}{2})), \text{ where } Y \text{ is a R.V.}$$

$$= E(e^{Yt} \cdot e^{\frac{\tau^2 t^2}{2}})$$

$$= e^{\frac{\tau^2 t^2}{2}} E(e^{Yt})$$

$$= e^{\frac{\tau^2 t^2}{2}} \cdot e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

$$= e^{\frac{t(\tau^2 + \sigma^2)}{2} + \mu t}$$

$$\therefore X \sim N(\mu, \tau^2 + \sigma^2)$$